

One Wrong Move: Target Clarity, Free Play, Crosswords

Report 010 for the target clarity and Three-Set Free Play revision. Generated May 3, 2026.

Final Response Summary

Implemented a clearer target model, added a second three-puzzle Free Play mode, introduced crossword-like and multi-click puzzle types, expanded validation, updated the lab, and captured the new visible UI states. Ladder Run remains the primary mode: one wrong committed move or timeout ends the run. Three-Set Free Play lets players solve three puzzles with recoverable mistakes and lower-is-better time scoring.

Repo State Summary

Repository	/Users/kevinhegg/Desktop/one-wrong-move
Feature branch	target-clarity-freeplay-crosswords
Starting main commit	87fbe70 Merge survival mode expansion
Remote	origin git@github.com:KevinHegg/one-wrong-move.git
Nested repo check	Only ./git found
Main branch status in this report	Feature branch validated and report generated; final merge and push occur after this PDF is committed.

Git Commands Run

```
pwd
git status --short
git branch --show-current
git log --oneline -10
git remote -v
find . -maxdepth 3 -name .git -type d
ls -la
git switch -c target-clarity-freeplay-crosswords || git switch target-clarity-freeplay-crosswords
node --check public/config.js
node --check public/symbols.js
node --check public/scoring.js
node --check public/puzzles.js
node --check public/game.js
node --check public/lab.js
node scripts/validate-scoring.js
node scripts/validate-puzzles.js
node scripts/validate-pattern-systems.js
node scripts/validate-survival.js
node scripts/validate-targeting.js
node scripts/validate-freeplay.js
node scripts/validate-word-puzzles.js
git diff --check
```

Files Created Or Changed

- README.md
- docs/game-design.md
- docs/pattern-systems.md
- index.html

- public/config.js
- public/game.js
- public/index.html
- public/lab.html
- public/lab.js
- public/puzzles.js
- public/style.css
- scripts/validate-pattern-systems.js
- scripts/validate-puzzles.js
- scripts/validate-survival.js
- scripts/validate-targeting.js
- scripts/validate-freeplay.js
- scripts/validate-word-puzzles.js
- docs/codex-reports/screenshots/010-*.png
- docs/codex-reports/screenshots/010-browser-verification.json
- docs/codex-reports/screenshots/010-row-target-check.json
- docs/codex-reports/010-target-clarity-freeplay-crosswords-report.pdf

Target Ambiguity Problem

Some puzzles described a row, equation, or output but rendered the whole board as tappable cells. That made a player wonder whether to tap the row, the broken die, the total, or any representative square. In Ladder Run, that ambiguity was especially punishing because one wrong committed move ends the run.

Targeting Model

Target type	Instruction	Behavior
cell	Tap the wrong tile/card/symbol/move.	Only exact candidate cells are enabled.
row	Tap the row that breaks the rule.	Any cell in the answer row is accepted.
column	Tap the column that breaks the rule.	Any cell in the answer column is accepted.
outputCell	Tap the wrong output.	Only output cells are enabled; inputs/gates/equals are disabled.
multiSelect	Select exact squares, then submit.	Selection clicks are planning; Submit commits.
twoStep	Make a two-step move, then submit.	Both steps are tentative until Submit.

Ladder Run Behavior

- Primary mode remains Ladder Run.
- One wrong committed move ends the run.
- Timeout ends the run.
- Correct answers pause timers and show feedback before Next Level.
- MultiSelect and twoStep selections are planning clicks; wrong Submit ends the run.

Three-Set Free Play

- Intro now offers Start Ladder Run and Three-Set Free Play.
- Free Play selects exactly three deterministic puzzles.
- Wrong attempts increment mistakes, show a hint, and keep the puzzle active.
- Correct answers pause the timer and advance to the next puzzle.
- After three solved puzzles, Free Play shows a lower-is-better score.

Free Play Scoring Formula

```
baseSeconds = Math.ceil(totalActiveMs / 1000)
mistakePenaltySeconds = mistakes * 10
scoreSeconds = baseSeconds + mistakePenaltySeconds
```

Crossword-Like Puzzles

- Mini Crossword Fill: choose one blank board square and one rack letter that complete both crossing words.
- Crossword Pair: select two blanks and two rack letters that repair a paired crossing system.
- Scrabble Cross remains a two-step word tile puzzle with separated rack tiles and curated common words.

More Multi-Click Puzzles

- Circuit Switch Pair: select the exact two switches that turn on the target light.
- Maze Bridge Repair: select the exact two broken bridge tiles that reconnect S to E.
- Existing Go Liberties, Yahtzee Fix, Maze Key Exit, and Tetris Fit continue to use multiSelect or twoStep commitment.

Validation Results

```
Validated lower-is-better legacy scoring and Survival Run scoring cases.
Validated 4500 daily rounds across 300 dates and 30 puzzle types.
Validated advanced pattern systems: 30 total types, 27 active, 300 dates, 5 attempts.
Validated Survival Run: 300 dates, 5 attempts, first 50 levels, answer modes, config, scoring, and puzzle-specific constraints.
Validated target clarity metadata, output-only targets, row targets, disabled cells, and title alignment.
Validated Three-Set Free Play generation, scoring, wrong-attempt behavior, share text, and default timer policy.
Validated crossword and word puzzle uniqueness metadata, curated words, rack separation, and accessibility labels.
git diff --check passed.
```

Browser Verification Results

- Intro shows both Ladder Run and Three-Set Free Play.
- Ladder briefing and Free Play briefing both hide the board until start.
- Ladder wrong move ends the run.
- Free Play wrong attempt increments mistakes and keeps playing.
- Free Play completes after exactly three puzzles and shows the lower-is-better score breakdown.
- Logic Gate Row enables only output targets.
- Dice Sum target behavior is unambiguous.
- Row-target puzzle accepted a non-answer cell in the answer row.
- Mini Crossword Fill, Crossword Pair, Circuit Switch Pair, and Maze Bridge Repair render correctly.
- Puzzle lab shows targeting metadata, Free Play preview, and hidden/shown answers.
- In-app browser console error log was empty.

Local Run Instructions

```
python3 -m http.server 8080 -d public
# Open http://localhost:8080
# Open http://localhost:8080/lab.html for the lab
# Use ?mode=freeplay or ?mode=ladder for mode testing
```

Lab Instructions

Use public/lab.html to inspect active production and retired puzzle types. The lab shows answer mode, target type, clickable and disabled counts, answer hidden/shown, break signature, evidence, Ladder stream preview, and Three-Set Free Play preview.

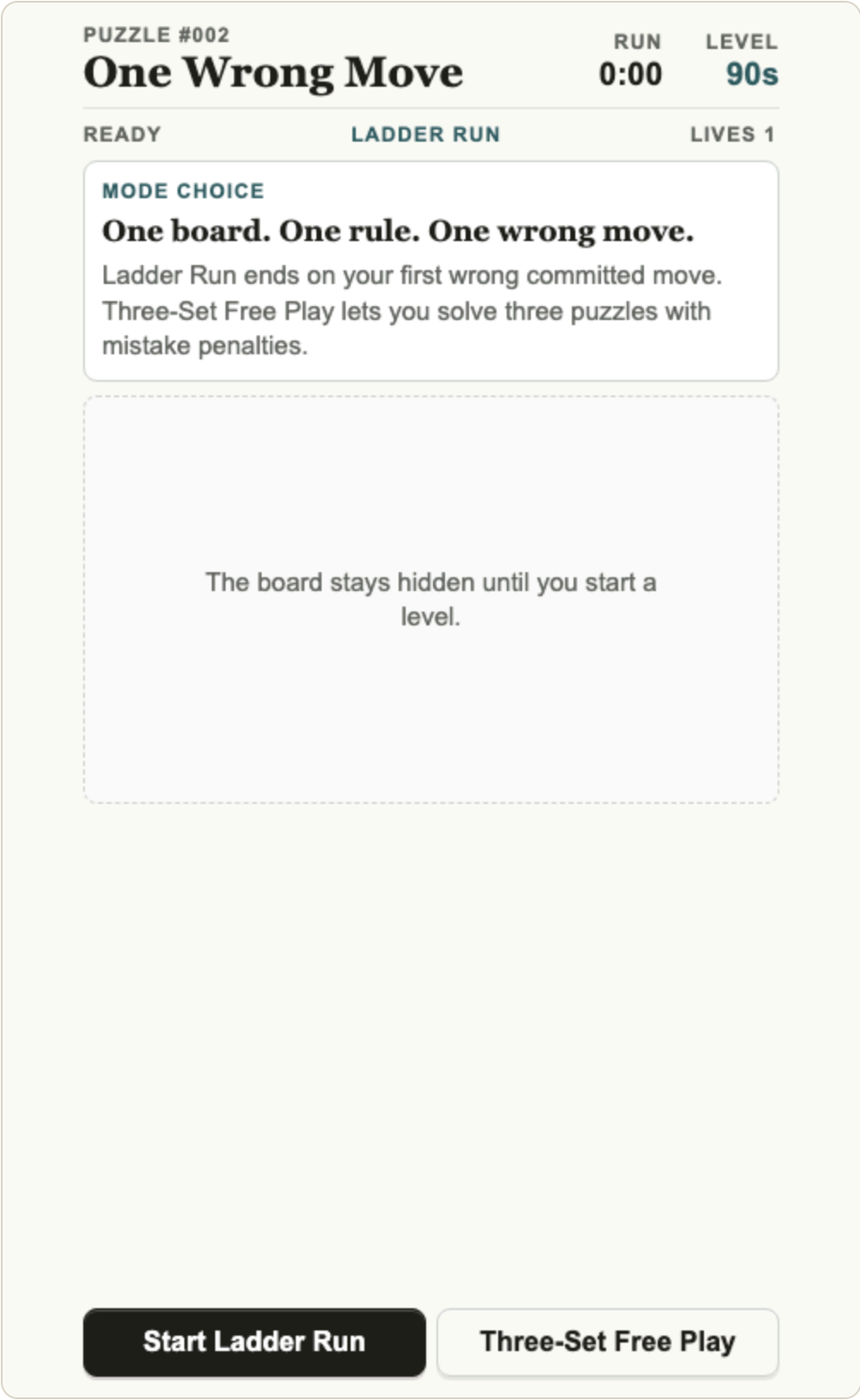
Netlify Deploy Settings

Build command	Leave blank
Publish directory	public

Screen Captures

Intro mode choice

010-intro-mode-choice.png



Ladder briefing

010-ladder-briefing.png

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL90s

LEVEL 1

LADDER RUN

LIVES 1

LEVEL 1: PAIR PACT · RELATIONSHIPS

Symbols travel in real-world partner pairs.
Tap the symbol with the wrong partner.

Tap one square. Timer and countdown are paused until
Start Level.

SYMBOLS:













BE

FL

EXAMPLE











Moon pairs with sun; key pairs with lock.

Level 1 is hidden until you press Start
Level.

Start Level

Restart

Ladder active board

010-ladder-active.png

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL90s

LEVEL 1

LADDER RUN

LIVES 1

LEVEL 1: PAIR PACT · RELATIONSHIPS

Symbols travel in real-world partner pairs.
Tap the symbol with the wrong partner.

Tap the one move that breaks the rule. One wrong move ends the run.

			BE	FL
BE	FL			

Restart

Ladder wrong move run-over

010-ladder-run-over-wrong.png

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL90s

RUN OVER

LADDER RUN

WRONG MOVES 1/1

RUN ENDED

0 levels in 0:00

Wrong move on Level 1. More levels is better; ties go to faster time.

Wrong move

Look for pairs that repeat elsewhere; one partner is with the wrong mate.

				
			BE	FL
				
BE	FL			
				

LEVELS0

RUN TIME0:00

Free Play briefing

010-freeplay-briefing.png

PUZZLE #002

RUN

0:00

LEVEL

—

PUZZLE 1 OF 3

FREE PLAY

MISTAKES 0

PUZZLE 1: DICE SUM · DICE

Rows show three dice and a total. Tap the die or total that breaks the sum.

Tap the wrong output. Timer and countdown are paused until Start Puzzle.

SYMBOLS:

1

2

3

4

5

6

EXAMPLE

2

3

1

=

6

2 + 3 + 1 = 6.

Level 1 is hidden until you press Start Level.

Start Puzzle

Restart

file:///tmp/owm-report-010.html

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Free Play active board

010-freeplay-active.png

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 1 OF 3

FREE PLAY

MISTAKES 0

PUZZLE 1: DICE SUM · DICE

Rows show three dice and a total. Tap the die or total that breaks the sum.

Only total cells are clickable. A wrong attempt adds a mistake; keep solving.

2	2	3	=	8
3	4	4	=	11
4	6	5	=	15
5	2	6	=	13
6	4	1	=	11

Restart

Free Play wrong attempt keeps playing

010-freeplay-wrong-keeps-playing.png

PUZZLE #002

RUN0:00

LEVEL—

One Wrong Move

PUZZLE 1 OF 3

FREE PLAY

MISTAKES 1

PUZZLE 1: DICE SUM · DICE

Rows show three dice and a total. Tap the die or total that breaks the sum.

Only total cells are clickable. A wrong attempt adds a mistake; keep solving.

Try again

Only totals can be wrong in this puzzle: add the three dice, then tap the bad total.

2	2	3	=	8
3	4	4	=	11
4	6	5	=	15
5	2	6	=	13
6	4	1	=	11

Restart

Free Play completed score screen

010-freeplay-results.png

PUZZLE #002

One Wrong Move

RUN0:01

LEVEL—

COMPLETE

FREE PLAY

MISTAKES 1

THREE-SET FREE PLAY COMPLETE

Solved 3 of 3

Lower is better: rounded-up solving time plus 10 seconds per mistake.

1	0	AND	=	0
0	0	OR	=	1
1	1	XOR	=	0
0	1	NAND	=	1
1	0	AND	=	0

LEVELS3 of 3

ENDED ON2s

RUN TIME0:01

RESULT12s

MORE LEVELS IS BETTER. TIES GO TO FASTER

SUM

Logic Gate Row output-only target screen

010-logic-output-targets.png

PUZZLE #002

One Wrong Move

RUN0:01

LEVEL—

PUZZLE 3 OF 3

FREE PLAY

MISTAKES 1

PUZZLE 3: LOGIC GATE ROW · LOGIC

Each row is A, B, gate, equals, output. Tap the wrong output.

Only output cells are clickable. A wrong attempt adds a mistake; keep solving.

1	0	AND	=	0
0	0	OR	=	1
1	1	XOR	=	0
0	1	NAND	=	1
1	0	AND	=	0

Restart

Logic Gate Row correct feedback

010-logic-correct-feedback.png

PUZZLE #002

One Wrong Move

RUN0:01

LEVEL—

PUZZLE 3 OF 3

FREE PLAY

MISTAKES 1

PUZZLE 3: LOGIC GATE ROW · LOGIC

Each row is A, B, gate, equals, output. Tap the wrong output.

Only output cells are clickable. A wrong attempt adds a mistake; keep solving.

Correct

That output is wrong for the gate.

1	0	AND	=	0
0	0	OR	=	1
1	1	XOR	=	0
0	1	NAND	=	1
1	0	AND	=	0

See Results

Restart

Dice Sum unambiguous target screen

010-dice-unambiguous-target.png

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 1 OF 3

FREE PLAY

MISTAKES 0

PUZZLE 1: DICE SUM · DICE

Rows show three dice and a total. Tap the die or total that breaks the sum.

Only total cells are clickable. A wrong attempt adds a mistake; keep solving.

2	2	3	=	8
3	4	4	=	11
4	6	5	=	15
5	2	6	=	13
6	4	1	=	11

Restart

Row-level target affordance

010-row-target-affordance.png

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 3 OF 3

FREE PLAY · VARIANT 17

MISTAKES 0

PUZZLE 3: ROW RHYTHM · LOGIC GRID

Four rows follow the same rhythm. Tap any cell in the row that breaks it.

Any cell in the broken row counts. A wrong attempt adds a mistake; keep solving.

A	B	C	D	E
B	C	D	E	A
C	D	B	A	B
D	E	A	B	C
E	A	B	C	D

Restart

Mini Crossword Fill briefing

010-mini-crossword-briefing.png

PUZZLE #002

RUN0:00

LEVEL—

One Wrong Move

PUZZLE 3 OF 3

FREE PLAY · VARIANT 5

MISTAKES 0

PUZZLE 3: MINI CROSSWORD FILL · CROSSWORD

Choose the empty crossword square, then choose the rack letter that makes both crossing words valid.

Make a two-step move, then Submit Move. Timer and countdown are paused until Start Puzzle.

SYMBOLS:

CATDOG SUN TREE A O E R

EXAMPLE

C □ T + A

One blank square plus one rack letter completes two common crossing words.

Level 3 is hidden until you press Start Level.

Start Puzzle

Restart

Mini Crossword Fill active board with rack

010-mini-crossword-active.png

PUZZLE #002

One Wrong Move

RUN0:01

LEVEL—

PUZZLE 3 OF 3

FREE PLAY · VARIANT 5

MISTAKES 0

PUZZLE 3: MINI CROSSWORD FILL · CROSSWORD

Choose the empty crossword square, then choose the rack letter that makes both crossing words valid.

Choose both parts of the move, then commit. A wrong attempt adds a mistake; keep solving.

			H	
		B		E
A	O	E	R	N

Submit Move

Clear selection

Restart

Mini Crossword Fill selected two-step move

010-mini-crossword-selected.png

PUZZLE #002

RUN0:01

LEVEL—

PUZZLE 3 OF 3

FREE PLAY · VARIANT 5

MISTAKES 0

PUZZLE 3: MINI CROSSWORD FILL · CROSSWORD

Choose the empty crossword square, then choose the rack letter that makes both crossing words valid.

Choose both parts of the move, then commit. A wrong attempt adds a mistake; keep solving.

			H	
		B		E
A	O	E	R	N

Submit Move

Clear selection

Restart

Crossword Pair active board

010-crossword-pair.png

PUZZLE #002

One Wrong Move

RUN

0:00

LEVEL

—

PUZZLE 2 OF 3

FREE PLAY · VARIANT 12

MISTAKES 0

PUZZLE 2: CROSSWORD PAIR · CROSSWORD

Fill two blanks so every crossing word is valid. Select the two blanks and two rack letters, then submit.

Plan by selecting squares, then commit. A wrong attempt adds a mistake; keep solving.

■	■	■	■	■
■	■	■	K	■
L	□	C	K	■
■	■	■	□	■
O	A	E	R	N

Submit Move

Clear selection

Restart

Circuit Switch Pair active board

010-circuit-switch-pair.png

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 3 OF 3

FREE PLAY · VARIANT 8

MISTAKES 0

PUZZLE 3: CIRCUIT SWITCH PAIR · CIRCUITS

Select the exact two switches to flip so the target light turns on.

Plan by selecting squares, then commit. A wrong attempt adds a mistake; keep solving.

1	1	XOR	AND	OFF
0	0			

Submit Move

Clear selection

Restart

Maze Bridge Repair active board

010-maze-bridge-repair.png

PUZZLE #002

One Wrong Move

RUN0:00

LEVEL—

PUZZLE 2 OF 3

FREE PLAY · VARIANT 5

MISTAKES 0

PUZZLE 2: MAZE BRIDGE REPAIR · MAZE

Select the two broken bridge tiles that repair the route from S to F.

Plan by selecting squares, then commit. A wrong attempt adds a mistake; keep solving.

.	.	×	.	S
.	■	×	×	■
.	■	■	×	■
.	■	■	×	■
F EXIT	■	■	■	■

Submit Move

Clear selection

Restart

Puzzle lab targeting metadata

010-lab-targeting.png

DEVELOPMENT LAB

One Wrong Move Lab

Game

DATE / SEED

2026-05-02

SESSION ATTEMPT

1

LEVEL LIMIT

60

Regenerate

Ladder stream preview

First 20 generated levels for this seed. Limit: 60s/level.

1. Dice Sum · Dice · identifyOne2. Mirror Trap · Relationships · identifyOne3. Latin Trap · Symbol Grammar · identifyOne4. Checkers Jump · Checkers · identifyOne5. Crossword Pair · Crossword · multiSelect6. Latin Trap · Symbol Grammar · identifyOne7. Maze Key Exit · Maze · twoStep8. Row Rhythm · Logic Grid · identifyOne9. Yahtzee Fix · Yahtzee · multiSelect10. Chess Attack · Chess · identifyOne11. Circuit Switch Pair · Circuits · multiSelect12. Go Capture Max · Go · chooseOne13. Chess Attack · Chess · identifyOne14. Go Liberties · Go · multiSelect15. Train Route · Routes · identifyOne16. Tetris Fit · Tetris · multiSelect17. Go Capture Max · Go · chooseOne18. Train Route · Routes · identifyOne19. Go Liberties · Go · multiSelect20. Maze Key Exit · Maze · twoStep

Three-Set Free Play preview

Three puzzles. Wrong attempts add mistakes. Score = ceil(active seconds) + mistakes × 10s.

1. Domino Chain · Dominoes · identifyOne · cell2. Scrabble Cross · Words · twoStep · twoStep3. Row Rhythm · Logic Grid · identifyOne · row

Production puzzle types (27)

CARDS · DIFFICULTY 1 · IDENTIFYONE

Show answer

Suit Cycle

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

A♠2♥3♦4♣7♥9♠J♣

Q♦

A♥A♦A♣A♠A♥

3♦3♦3♠3♥3♦

RELATIONSHIPS · DIFFICULTY 1 · IDENTIFYONE

Show answer

Pair Pact

Symbols travel in real-world partner pairs. Tap the symbol with the wrong partner.

☾☀☷☱☲☳BE

FL

☠☙•☺☛

☾☀•☺☛

DOMINOES · DIFFICULTY 1 · IDENTIFYONE

Show answer

Domino Chain

Neighboring domino halves must match. Tap the domino that breaks the chain.

123456

4|55|66|11|22|3

5|66|11|22|33|4

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Puzzle lab Free Play 3-set preview

010-lab-freeplay-preview.png

Three-Set Free Play preview

Three puzzles. Wrong attempts add mistakes. Score = ceil(active seconds) + mistakes × 10s.

1. Domino Chain · Dominoes · identifyOne · cell2. Scrabble Cross · Words · twoStep · twoStep3. Row Rhythm · Logic Grid · identifyOne · row

Production puzzle types (27)

CARDS ·
DIFFICULTY 1 ·
IDENTIFYONE

Show answer

Suit Cycle

Suits cycle across the row and shift by row. Tap the card with the wrong suit.

A♠ 2♥ 3♦ 4♣ 7♥ 9♣ J♠

Q♦

A♥ A♦ A♣ A♠ A♥

3♦ 3♦ 3♣ 3♥ 3♦

5♣ 5♣ 5♥ 5♦ 5♣

7♣ 7♥ 7♦ 7♣ 7♣

9♥ 9♦ 9♣ 9♣ 9♥

Break signature

suit-cycle|duplicate-suit|6|3+3{"rank":"3","suit":"diamond","suitIndex":1}

Target type

cell

Clickable targets

25 clickable · 0 disabled

Broad target

Answer cells: 7

Evidence

The rows shift a four-suit cycle; the tapped card uses the wrong suit for its position.

Answer

RELATIONSHIPS ·
DIFFICULTY 1 ·
IDENTIFYONE

Show answer

Pair Pact

Symbols travel in real-world partner pairs. Tap the symbol with the wrong partner.

☾ ☀ 📖 📱 🏠 📏 BE

FL

☠ 📏 . 📖 📱

☾ ☀ . 📖 📱

👁 👁 . BE FL

☠ 📏 . 👁 👁

BE FL . ☾ ☾

Break signature

pair-pact|leader-repeated|24|☾☾

Target type

cell

Clickable targets

25 clickable · 0 disabled

Broad target

Answer cells: 25

Evidence

The board repeats real-world pair pacts; the tapped symbol is not the established partner.

Answer

DOMINOES ·
DIFFICULTY 1 ·
IDENTIFYONE

Show answer

Domino Chain

Neighboring domino halves must match. Tap the domino that breaks the chain.

1 2 3 4 5 6

4|5 5|6 6|1 1|2 2|3

5|6 6|1 1|2 2|3 3|4

6|1 1|2 2|3 3|4 4|5

1|2 2|3 3|4 4|2 5|6

2|3 3|4 4|5 5|6 6|1

Break signature

domino-chain|right-side-mismatch|18|4|2{"left":4,"right":2}

Target type

cell

Clickable targets

25 clickable · 0 disabled

Broad target

Answer cells: 19

Evidence

Neighboring domino halves match across every clean row; the tapped domino breaks that handoff.

Answer

file:///tmp/owm-report-010.html

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Puzzle lab answer hidden

010-lab-answer-hidden.png

Production puzzle types (1)

CROSSWORD · DIFFICULTY 2 · TWOSTEP

Mini Crossword Fill

Choose the empty crossword square, then choose the rack letter that makes both crossing words valid.

CAT DOG SUN TREE A O E R

H

B

E

A

O

E

R

N

Break signature

mini-crossword-fill|bee-hive-e|18.23|E

Target type

twoStep

Clickable targets

6 clickable · 19 disabled

Broad target

Answer cells: 19, 24

Evidence

Only one square-letter pair makes BEE and HIVE.

Steps

board-square: cell 19, rack-tile: cell 24

Answer

Hidden

Validator

valid · 18, 23

Retired / lab-only puzzle types (0)

No puzzle types match this filter.

Puzzle lab answer shown

010-lab-answer-shown.png

Production puzzle types (1)

CROSSWORD · DIFFICULTY 2 · TWOSTEP

Mini Crossword Fill

Choose the empty crossword square, then choose the rack letter that makes both crossing words valid.

CAT DOG SUN TREE A O E R

H

B

✓

E

A

O

E

R

✓

N

Break signature

mini-crossword-fill|bee-hive-e|18.23|E

Target type

twoStep

Clickable targets

6 clickable · 19 disabled

Broad target

Answer cells: 19, 24

Evidence

Only one square-letter pair makes BEE and HIVE.

Steps

board-square: cell 19, rack-tile: cell 24

Answer

Cells 19, 24: That square and rack letter complete both crossing words.

Validator

valid · 18, 23

Retired / lab-only puzzle types (0)

No puzzle types match this filter.